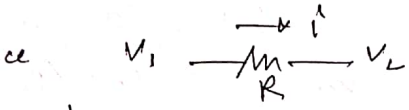
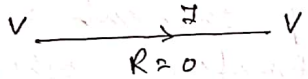
1 Reference 

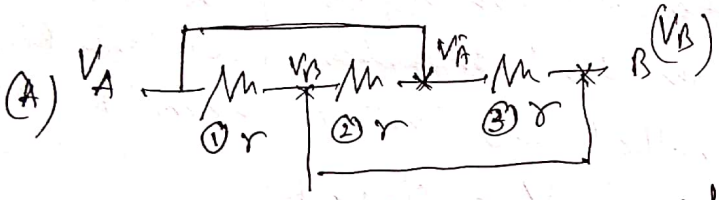
if $R = 0$,

$V_1 - V_2 = i \times 0 = 0 \quad \therefore V_1 = V_2$

So on a normal plain wire of almost zero resistance potential is same at all pts.

ie 

Now!

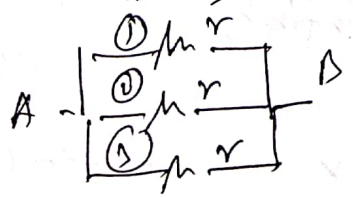


{ Let potential at A be V_A & that at B be V_B }

on the zero resistance wire connected at pt A & at the jn betn resistance ① & ③, potential on both sides are V_A.

Same is for other zero resistance wire connected betn pts B & at jn betn. resistor ① & ②, on both sides or its potential is V_B.

Hence, Note that across every resistance r①, r②, r③, same p.d $V_A - V_B$ exists. So they are in parallel combination

$\equiv$ 

$\therefore \frac{1}{R_{AB}} = \frac{1}{r} + \frac{1}{r} + \frac{1}{r} = \frac{3}{r}$

$\boxed{R_{AB} = r/3}$

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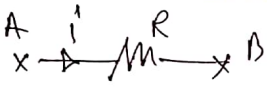
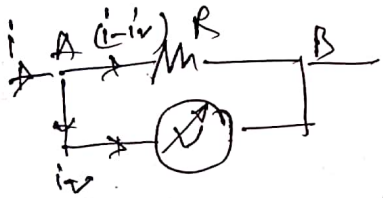
(2)



if current 'i' flows thro' resistance 'R' then p.d across R is $V_A - V_B = iR$. — (1)

To measure this p.d we connect a voltmeter across R in parallel combination betn. pts A & B.



Then a part of current i, i_v flows thro' voltmeter (otherwise its pointer will not deflect) & remaining $(i - i_v)$ flows thro' ~~voltmeter~~ resistance 'R'.

∵ R & V are in parallel so voltage across 'R' measured by voltmeter is now $V = (i - i_v)R$

$$\boxed{V = iR - i_vR} \quad \text{— (2)}$$

iR = correct voltage across R to be measured

i_vR = error in measurement by voltmeter.

error in measurement by voltmeter is i_vR. This error = 0 if $i_v = 0$, ie if voltmeter draws no current.

But this is an ideal condition, possible only when resistance of voltmeter is infinity (∞), But then voltmeter will show no deflection/reading. In practice this error is minimised by reducing i_v, ie increasing Resistance of voltmeter.

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